Directions: Select **3** topics. For the chosen topics,

- Review all the material and answer the requested questions to the best of your ability. Use your textbook, classmates, or any available internet resources to ensure you've found the correct answer.
- These concepts may not have been discussed in class and you should seek alternate resources to learn about the context for the problem. **Bolded terms** may help you find solution methods.
- Explain your solutions using full sentences as if you were creating a guide for a classmate who has all the linear algebra information you have but has not researched the secondary topic. Explain new concepts and why they are relevant to answering the questions.
- You can (and should!) work with classmates and develop the same solutions as others in class. However, your write up must not be identical to anyone else's submitted work or no credit will be given.

Deadline for turning these in is 11/13/26 at 2:30 pm. Please turn in on paper in class or submit to my office mailbox.

1. Fourier Analysis

   A. For vectors in $\mathbb{R}^n$, ***orthogonal*** basis vectors are useful. The standard basis in $\mathbb{R}^n$ is orthogonal, we can check by computing the ***inner product*** of the vectors in the basis. The inner product for $\mathbb{R}^n$ is the dot product, which is 0 when computed for orthogonal vectors. Now consider the inner product for continuous functions on the domain $[-\pi, \pi]$,

   $$\langle f, g\rangle = \int_{-\pi}^{\pi} f(x)g(x)dx$$

   Compute the inner products

   i. $\langle 1, \cos(x)\rangle$
   ii. $\langle 1, \sin(x)\rangle$
   iii. $\langle \sin(x), \cos(x)\rangle$

   B. The infinite set

   $$B = \{1,\ \cos(x),\ \sin(x),\ \cos(2x),\ \sin(2x),\ \ldots\}$$

   is an orthogonal basis for a vector space of periodic functions. That is, any periodic function $f(x)$ can be represented as

   $$f(x) = a_0 + a_1\cos(x) + b_1\sin(x) + a_2\cos(2x) + b_2\sin(2x) + \ldots$$

   .

   i. Why would you want an infinite basis for this vector space?
   ii. Suppose you have observed data that follows a periodic trend. Give two reasons it would be useful to store the data as a Fourier series rather than in a standard data table.
   iii. What are the drawbacks of having an infinite basis for approximation?

   C. A coastal scientist models the water height at a harbor by

   $$h(t) = 2.4 + 1.8\cos\left(\frac{2\pi t}{12}\right) + 0.5\cos\left(\frac{2\pi t}{24}\right)$$

   where t is measured in hours.

   i. Identify the basis functions appearing in the model. What do they represent in this model?
   ii. What are the coordinates of $h(t)$?
   iii. What does the constant term 2.4 represent physically?
   iv. Which oscillation has the greatest influence on the water level?

   D. A sound engineer records a musical note and models the waveform as

   $$s(t) = 3.5\sin(440t) + 1.2\sin(880t) + 0.3\sin(1320t) + 0.05\sin(1760t)$$

   i. Which frequency contributes most to the sound?
   ii. The engineer wants to compress the sound file by discarding all coefficients with a magnitude less than 0.1. What is the new approximation? Why will this approximation still sound like the original?

2. Principal Component Analysis

   A. Consider the standardized test scores of these five students,

	SAT math	SAT reading
A	520	510
B	580	590
C	610	620
D	680	690
E	740	730

      i. Write each data point as a vector in $\mathbb{R}^2$.
      ii. Plot the data points and describe their pattern.
      iii. Do the points appear to spread equally in all directions?
      iv. Plot the alternate set of axes
$$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$
      Which axis aligns best with the data?
      v. Find the coordinates of each data point with respect to B. What do you notice about the coordinates corresponding to $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$?
      vi. Suppose one axis in B is interpreted to correspond with "Overall Standardized Test Performance" and the other corresponds to "Difference Between Math and Reading Performance". Which is which? Are these new axes more useful to a college admissions officer?
      vii. If you wanted to identify each data point by just one coordinate, which axis would you choose?

   B. Suppose you have a dataset with information vectors in $\mathbb{R}^{20}$ (twenty measurements for each sample). After performing PCA, you discover nearly all the variation is captured by just two new basis vectors.
      i. How many coordinates were required before PCA and how many are required after?
      ii. What advantages are there to describing the data by just the values associated with the two new basis vectors?

   C. Consider the data of 5 users of a streaming service, The analysts working with the data perform

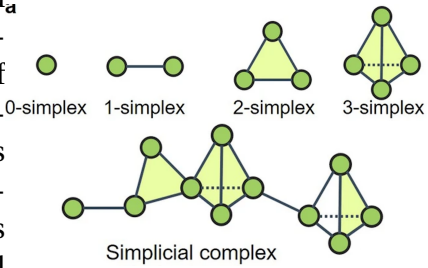
	Hours Watching TV	Hours Watching Movies
A	2	3
B	4	4
C	6	7
D	8	9
E	10	11

   PCA and identify their new basis as representing "Total Viewing Activity" and "Preference for TV vs Movies".

i. Which coordinate would contain more variation?
ii. Which would help identify heavy users of the streaming service?
iii. Which would help identify users with unusual viewing habits?
iv. Do the new coordinates provide more useful information than the original measurements?

3. Topological Data Analysis

A. Consider data points that have some form of connecting, for example cities that are connected by train routes. Each data point can be assigned a letter, let's consider points A, B, C, and D. Each data point is a 0-simplex, labeled by it's letter. Each connection is a 1-simplex, labeled by the two letters it connects. If three points are connected, they form a filled triangle called a 2-simplex with 0-simplices as vertices and 1-simplices as as edges that is labeled by the points at its vertices. If the fourth is connected, it is a tetrahedron called a 3-simplex with the 2-simplices as triangular faces and is also labeled by its vertices. The set of all the vertices and their connections is called a ***simplicial complex***.



   i. Let K be a tetrahedron where the points A, B, C and D are all connected. How many 0-simplices are in K? How many 1-simplices? How many 2-simplices? How many 3-simplices?

B. Simplices can be connected together to form ***chains***. A chain of 0-simplex objects, in our example, is a list of cities to be considered. They can be represented by a linear combination

$$\alpha A + \beta B + \gamma C + \delta D$$

where the coefficients could theoretically represent the relative importance of each city. The set of all of these linear combinations is denoted $C_0(K)$. A chain of 1-simplex objects, in our example, is a path of train routes connecting the cities. If there's a route from City A to City B and from City B to City C and from City C to City D, this full route is denoted by the chain

$$AB + BC + CD$$

where, again, weights could be added to describe some relevant aspect of the routes. The set of all of these linear combinations is denoted $C_1(K)$.

   i. Given K from above, find a basis for $C_0(K)$, $C_1(K)$, $C_2(K)$, and $C_3(K)$.
   ii. What does the dimension of the basis count in each example?
   iii. Why would it be inconvenient to combine elements from C_0 and C_1?

C. We can discuss the routes using a ***linear transformation*** map defined

$$\partial_1 : C_1 K \to C_0 k$$

where

$$\partial_1(AB) = B - A.$$

Recall that linear transformations can be represented by multiplying a vector by a matrix, so the properties of linearity hold. Thus

$$\partial_1(AB + BC) = \partial_1(AB) + \partial_1(BC)$$

   i. Compute $\partial_1(AB + BC)$ *(get a result in $C_0(K)$).*

ii. Compute $\partial_1(AB+BC-AC)$. What does this represent in the context the train route example?

iii. Construct the matrix of ∂_1 using the basis for C_0K and the basis for $C_1(K)$. Use the matrix to compute $\partial_1(AB+AC-BC)$.

D. Consider the simplicial complex J where the basis of $C_0(J) = \{A, B, C, D\}$ and the basis of $C_1J = \{AB, BC, CD, DA\}$.

i. Draw a diagram of J.

ii. Consider the chain $L = AB + BC + CD + DA$. Find $\partial_1(J)$. Considering the diagram of J, what is the significance of this chain? What feature of the simplicial complex has been discovered through linear algebra?

E. Eight sensors are placed around a forest. Two sensors are connected by an edge whenever their sensing regions overlap. A triangle is filled whenever three sensors overlap pairwise.

i. If a loop was found in this simplicial complex, what would it represent? Why is this information important?

ii. How would adding more sensors in the loop change the system?

4. Cryptography

   A. Consider the vector space $\mathbb{Z}_5$ where the vectors and the eligible scalars are $\{0,1,2,3,4\}$ and vector arithmetic is performed ***modulo 5***.

      i. Compute $3+4 \mod 5$
      ii. Compute $4+4 \mod 5$
      iii. Compute $3\cdot 4 \mod 5$
      iv. Find a value for x such that $2x \equiv 1 \mod 5$

   B. Let $V = \mathbb{Z}_5^2$ be the vector space of ***tuples*** in $\mathbb{Z}_5$. Sample vectors in V would be $(2,0)$ or $(1,4)$.

      i. How many vectors are in $\mathbb{Z}_5^2$?
      ii. Compute $(2,4)+(3,2)$.
      iii. Compute $3(1,4)$.

   C. Consider the vectors $\vec{\boldsymbol{v}}_1 = \begin{bmatrix}1\\0\end{bmatrix}$ and $\vec{\boldsymbol{v}}_2 = \begin{bmatrix}0\\1\end{bmatrix}$ in $\mathbb{Z}_5^2$.

      i. Show $\vec{\boldsymbol{v}}_1$ and $\vec{\boldsymbol{v}}_2$ are linearly independent in $\mathbb{Z}_5^2$. Do they form a basis? Explain.
      ii. Find the coordinates of $(3,4)$ relative to $\vec{\boldsymbol{v}}_1$ and $\vec{\boldsymbol{v}}_2$.
      iii. Find the coordinates of $(3,4)$ relative to $\vec{\boldsymbol{u}}_1 = \begin{bmatrix}1\\1\end{bmatrix}$ and $\vec{\boldsymbol{u}}_2 = \begin{bmatrix}1\\2\end{bmatrix}$. How do these compare to the coordinates with basis $\vec{\boldsymbol{v}}_1$ and $\vec{\boldsymbol{v}}_2$?
      iv. Why would changing bases in this system be useful for a company trying to protect their data?

   D. Consider the linear transformation

$$T(x) = \begin{bmatrix}1 & 2\\ 3 & 1\end{bmatrix}\vec{\boldsymbol{x}}$$

      i. Encode the messages $\begin{bmatrix}1\\2\end{bmatrix}$ and $\begin{bmatrix}3\\4\end{bmatrix}$. What must be true of A if we need to recover the original messages?

   E. Consider the cryptographic system using vectors from $\mathbb{Z}_{256}^{100}$

      i. How many coordinates describe each message?
      ii. Why is $\mathbb{Z}_{257}$ better than $\mathbb{Z}_{256}$?
      iii. Describe the role of vector spaces and bases in this system.
      iv. Why must matrices be used in this type of encryption?

5. Operations Research

A. A small furniture company building desks and bookcases that sell for \$300 and \$200 respectively. It takes 1 hour to do the final construction of each object and requires 2 units of prepared parts for a desk but only one 1 unit for the bookcase. If a given project has been allotted 10 hours and 14 units of prepared parts, the maximum profit obtainable is calculated by solving the linear programming problem

$$\begin{aligned} \text{Maximize} \quad & 3x+2y \\ \text{subject to} \quad & x+y \le 10 \\ & 2x+y \le 14 \\ & x,\ y \ge 0 \end{aligned}$$

It's important that the ***objective function*** $3x+2y$ is in the span of the ***constraints***.

i. If the objective function is written as the vector $\begin{bmatrix}3\\2\end{bmatrix}$, what are the vectors $\vec{c_1}$ and $\vec{c_2}$ that represent each of the two constraints?

ii. Is the objective function in the span of the constraints?

iii. Are the constraint vectors a basis for $\mathbb{R}^2$?

iv. What are the coordinates for the objective function relative to the constraint vectors? That is, find y_1 and y_2 that satisfy

$$\begin{bmatrix}3\\2\end{bmatrix} = y_1\vec{c_1} + y_2\vec{c_2}$$

These coordinates are called ***dual variables*** and represent ***shadow prices***. In particular, y_1 represents the value of 1 hour of labor and y_2 represents the value of 1 unit of prepared parts.

B. Changing vectors $\vec{c_1}$ and $\vec{c_2}$ back to the constraint expressions $x+y$ and $x+2y$ respectively shows the result from the previous section has form

$$3x+2y = y_1(x+y) + y_2(x+2y)$$

i. Notice the right hand side is a linear combination of the constraints with the dual variables as coefficients.

   A. Multiply both sides of the first constraint by y_1.

   B. Multiply both sides of the second constraint by y_2.

   C. Finally, sum the constraints to form the linear combination above.

   D. Use this result to give the upper bound of the objective function in terms of the dual variables.

ii. Rewrite the right hand side of this equation so that linear combinations of the dual variables are the coefficients of x and y.

   A. The result from the previous section still equals $3x+2y$. Use this fact to solve for the dual variables.

   B. What is the upper bound of the objective function subject to the given constraints? In terms of shadow prices, this upper bound is the total value of all available resources.

C. Because we were able to find unique solutions for the dual variables, the ***primal*** *(original)* problem had ***strong duality***. If there had been infinitely many solutions, the problem would have ***weak duality***. When we have weak duality, it may be easier to formulate the associated ***dual*** problem and look for the smallest possible upper bound.

i. Assuming we could not find unique values for y_1 and y_2, how would we write the upper bound of the primal problem? This is the dual objective function.

ii. In the primal problem, x and y represented physical properties and therefore could not be negative, so we had the constraint $x, y \geq 0$. In the dual problem, our dual variables cannot be negative or they will reverse the inequalities. So we must have as a constraint $y_1, y_2 \geq 0$.

iii. If the problem had weak duality, it would be possible for the results of $B.ii.A$ to be greater than their associated values of 3 and 2. Write out these last two constraints of the dual problem.

iv. Use the template

$$\begin{aligned} \text{Minimize} \quad & \alpha_1 y_1 + \alpha_2 y_2 \\ \text{subject to} \quad & \beta_1 y_1 + \beta_2 y_2 \geq \beta_3 \\ & \gamma_1 y_1 + \gamma_2 y_2 \geq \gamma_3 \\ & y_1,\ y_2 \geq 0 \end{aligned}$$

to formulate the dual problem. That is, use your previous work to identify

$$\{\alpha_1, \alpha_2, \beta_1, \beta_2, \beta_3, \gamma_1, \gamma_2, \gamma_3\}$$

and write the solution in the given form of a linear programming problem.

D. A shipping company transports freight using two shipping routes. Let x be the number of shipments using Route A and y be the number of shipments using Route B. The following linear programming problem shows how they can maximize their profit subject to certain capacity limits.

$$\begin{aligned} \text{Maximize} \quad & 8x + 6y \\ \text{subject to} \quad & 2x + y \leq 20 \\ & x + 3y \leq 24 \\ & y_1,\ y_2 \geq 0 \end{aligned}$$

i. Graph the constraints and identify the point in the ***feasible*** region that maximizes the objective function. What is the maximum attainable profit?

ii. Use the process in the furniture example to find and solve for the dual variables.

iii. Write the dual problem based on this work. Compare the ***objective value*** of the dual solution to the objective value of the primal solution.